

BayesCog: A freely available course in Bayesian statistics and hierarchical Bayesian modeling for psychological science

Lei Zhang^{1,2,3} and Aamir Sohail^{1,2}

¹ Centre for Human Brain Health, School of Psychology, University of Birmingham, Birmingham, UK ² Institute for Mental Health, School of Psychology, University of Birmingham, Birmingham, UK ³ Centre for Developmental Science, School of Psychology, University of Birmingham, Birmingham, UK

DOI: 10.xxxxxx/draft

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright¹⁴ and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).¹⁷

Summary

We present [BayesCog](#), an openly-available online course for the computational modeling of human behavior (i.e., cognitive modeling) using Bayesian inference, with reinforcement learning as a core example throughout the course. Assuming little to no prior experience, audience of this course will be formally grounded in key concepts including Bayesian statistics and reinforcement learning, and practically, will build, assess, compare, and validate models using the R interface to the Stan programming language, RStan. Starting with binary choice models, the audience will learn to estimate parameters representing latent components of behavior by fitting reinforcement learning models, both at the individual and group-level, eventually with hierarchical modeling.

The course is generally suitable for those interested in developing models of human cognition at any level of experience. In making the course openly available, we aim for computational modeling under the Bayesian approach to be more strongly represented in the psychological sciences.

Statement of Need

Computational modeling is a general framework that uses mathematical equations to infer unobserved latent processes, variables, and parameters from observed data. Whilst implemented in other disciplines (e.g., physics, chemistry, and, astronomy) for centuries, its application specifically towards understanding the human mind (i.e., learning, memory, decision-making, language) ([Farrell & Lewandowsky, 2018](#)) is a relatively recent approach (known as cognitive modeling), one exponentially increasing in popularity ([Palminteri et al., 2017](#)). By formalising cognitive processes as mathematical operations and free parameters, cognitive models generate specific, testable hypotheses about observable behavior, which can be objectively compared, verified, and falsified ([Guest & Martin, 2021](#); [Palminteri et al., 2017](#); [W. Pan et al., 2022](#); [Rocca & Yarkoni, 2021](#); [Zhang et al., 2020](#)). When combined with other modalities of measurement, such as functional magnetic resonance imaging (fMRI), cognitive models present a key framework for understanding how the brain implements cognitive processes such as decision making and (social) learning ([Eckstein et al., 2021](#); [FeldmanHall & Nassar, 2021](#)) and their aberration in mental health disorders ([Hauser et al., 2022](#); [Huys et al., 2016](#); [Sohail & Zhang, 2024](#)).

Complementing this approach is the application of Bayesian methods for parameter estimation ([Annis & Palmeri, 2018](#)), which applies the Bayes rule to obtain the posterior distribution of model parameters given the observed data (1):

$$P(\theta|D) = \frac{P(D|\theta) \cdot P(\theta)}{P(D)} \quad (1)$$

where $P(\theta|D)$ is the posterior distribution, $P(D|\theta)$ is the likelihood, $P(\theta)$ is the prior distribution, and $P(D)$ is the marginal likelihood.

Bayesian methods confer advantages over frequentist approaches (e.g., Maximum Likelihood Estimation, MLE), by quantifying the uncertainty, and when implemented hierarchically, permit simultaneous estimation of individual and group-level parameters while appropriately pooling information across participants (M. D. Lee, 2011). Historically restrictive due to their computational burden, these methods are now more accessible through the development of multiple programming languages and software such as JAGS (Plummer & others, 2003) and Stan (Carpenter et al., 2017) which optimize the sampling process used for parameter estimation using approaches such as Markov chain Monte Carlo (MCMC).

Using Bayesian models of cognition in one's own research requires a conceptual understanding of both Bayesian statistics and cognitive modeling, as well as the practical skills to translate these models into computer code. Textbooks (Kruschke, 2014; Lambert, 2018; M. D. Lee & Wagenmakers, 2014; McElreath, 2018) and tutorial papers (Baribault & Collins, 2023; Lockwood & Klein-Flügge, 2021; Wilson & Collins, 2019; Zhang et al., 2020) have made learning these skills more accessible, but are often not freely available, and challenging for researchers, especially early career researchers with little to no prior experience. Additionally, whilst free, online courses for the computational modeling of cognition currently exist, these are few and far between, and exclusively cover non-Bayesian implementations in Python (Rhoads & Gan, 2022) and MATLAB (O'Reilly & Ouden, 2015), as well as Bayesian approaches in Python (Niv Lab, 2021). A full course implementing Bayesian models of cognition through the open source R programming language is therefore a valuable yet currently non-existent resource.

Learning Objectives

In the course, students will:

- Build a foundational knowledge of Bayesian statistics and inference, and how it differs from frequentist definitions of probability
- Understand Bayesian parameter estimation and the conceptual basis of sampling techniques including Markov chain Monte Carlo (MCMC)
- Understand how Bayesian statistics can be applied to uncover latent processes and parameters of cognition through cognitive modeling
- Practically build and code discrete choice and regression models using RStan
- Learn the basic reinforcement learning (RL) concepts
- Build a simple RL model (Rescorla-Wagner) using RStan
- Understand the conceptual basis of hierarchical models and implement them practically in RStan
- Be able to evaluate model performance and perform model comparison using the `loo` package in R
- Ultimately be able to apply computational modeling in their own experiments

Prior to the first workshop, the course begins by summarizing the broader philosophy in which the techniques and methods are implemented. Specifically, this concerns Marr's influential three levels of analysis (Marr, 1982), which describe how algorithmic-level (as opposed to computational and implementation levels) models can help understand behavior. Doing so shapes the course material within this framework, demonstrating the necessity for building strong theories in psychology (Press et al., 2022). Assuming no prior experience with programming, the course properly (Workshop 01) begins with a

87 basic introduction to the R programming language and the RStudio interface (RStudio
88 Team, 2020). This introductory workshop first provides a general introduction to data
89 structures, variables, and packages, after which students will work with simulated data
90 from a reversal learning task, performing basic summary statistics including correlation
91 and regression, and visualizing the results using the ggplot2 package (Wickham, 2016).

92 In Workshop 02, students will be introduced to Bayesian statistics, learning the differences
93 between Bayesian and frequentist definitions of probability. This can be conceptually
94 difficult for those with little statistical or mathematical experience, the material therefore
95 slowly builds to the concept of Bayes' theorem from simpler probability concepts including
96 joint, conditional and marginal probability, and probability types – discrete and continuous.
97 These concepts are put into practical use in Workshop 03, where the Bayesian approach is
98 transformed from a purely mathematical concept to a system where one can determine the
99 values of unknown parameters from data. The goal of Bayesian inference - computing the
100 probability distribution of model parameters given the observed data – is also introduced,
101 firstly using a simple instance, grid approximation, adapted from (McElreath, 2018).
102 Students will subsequently learn that for more complex models, sampling procedures
103 which approximate the posterior distribution are used. To this end, a popular approach –
104 Markov chain Monte Carlo (MCMC) – is conceptually introduced.

105 At this point, students will have the conceptual knowledge of Bayesian statistics and
106 probability to practically implement models. Workshop 04 introduces students to the
107 Stan programming language (Carpenter et al., 2017) and it's R interface RStan (Stan
108 Development Team, 2024). Following an overview of Stan syntax, students will construct a
109 binomial model for a coin flipping toy example, estimating the same parameter calculated
110 by grid approximation in the previous workshop. In doing so, students will become
111 familiar with the relationship between Stan and R, with models being coded in a *.stan
112 file, and sourced to the respective R script. Students will also examine basic outputs
113 of the sampler, including the summary statistics, and by plotting the posterior density
114 distribution. Workshop 5 builds upon this introductory workshop by introducing some
115 specific properties and advantageous features of Stan (as opposed to other packages like
116 JAGs), including vectorization and variable declaration, and introduces two more models:
117 the Bernoulli model and linear regression.

118 Having built a solid foundation of understanding Bayesian statistics in Workshops 1-5, in
119 Workshop 6, we turn to the specific application of these methods to infer latent cognitive
120 processes. A basic overview to the principles of cognitive modeling is followed by an
121 introduction to reinforcement learning, a popular theory of human behavior that has been
122 widely used in the last decades (Daw et al., 2011; Dayan & Niv, 2008; D. Lee et al., 2012;
123 Niv, 2009). To this end, we introduce a simple reinforcement learning algorithm consisting
124 of the Rescorla-Wagner model (Rescorla & Wagner, 1972) that uses an error-driven rule
125 (e.g., through reward prediction error) to update value computation, and a softmax choice
126 rule quantifies the stochasticity and randomness in human action. Students will then
127 practically implement this 'Simple RL' model in Stan, for simulated choice data for a
128 single subject, before fitting multiple subjects. For the latter exercise, students will do so
129 in two ways, by firstly looping across subjects and fitting each subject individually. In
130 doing so, they will learn about statistical concerns with parameter estimation in groups
131 and drawbacks of these two approaches. Workshop 7 directly builds upon this topic by
132 introducing hierarchical Bayesian models (D. Lee et al., 2012) for simultaneously estimating
133 both group- and individual level parameters. Conceptually, this means that all individuals
134 are assumed to be drawn from a common, higher-level, group distribution, meanwhile
135 maintain their individual level variation. Introducing the concepts of 'hyperparameters'
136 and 'hyperpriors', students will then run a hierarchical implementation of the Simple RL
137 model. Given that Bayesian cognitive models often require troubleshooting for parameter
138 estimation (Baribault & Collins, 2023), optimization strategies are also introduced in this
139 workshop. Specifically, this covers Stan's sampling parameters and reparameterization;

140 the latter being particularly relevant for hierarchical models.

141 Model comparison is introduced in Workshop 8, with the basis behind model fitting,
142 predictive accuracy and information criterion firstly described. Taking Widely Applicable
143 Information Criterion (WAIC) as an example, students will subsequently compare two
144 RL models: the Simple RL (updating value computation only for the chosen option)
145 and ‘Fictitious RL’ (updating value computation for both the chosen and unchosen
146 options) using the `loo` package (Vehtari et al., 2015) in R, and plot posterior predictive
147 checks as a measure of model validation. The final workshop (Workshop 9) describes key
148 strategies for code writing styles and code debugging in Stan, using a purposely error-laden
149 delay-discounting model (M. D. Lee & Wagenmakers, 2014) for students to interactively
150 troubleshoot problematic code. To conclude, a published study (Crawley et al., 2020)
151 where computational models were implemented to uncover learning differences is described,
152 providing real-case examples of model and parameter recovery.

153 The course begins from the shared assumption of no-to-little prior experience among
154 students with programming and statistics, however specific workshops may be skipped for
155 certain audiences. For example, if students are comfortable with the RStudio environment,
156 teachers could begin with Workshop 2. Similarly, if looking to introduce experienced
157 Stan users to the specific topic of modeling cognitive processes through RL, Workshop
158 6 provides an ideal starting point. Any tutorial can be adopted, transformed, or built
159 upon for other educational purposes (e.g., courses, single class sessions, workshops) under
160 a Creative Commons Attribution-ShareAlike 4.0 International License.

161 All course workshops are accompanied by example data and scripts, the latter provided in
162 both uncompleted and completed versions where appropriate. All software required for
163 the course are open-source and easy to install; instructions are provided in the ‘Course
164 overview’ page. Whilst specific R packages (`rstan`, `loo`, `ggplot2`) are required, the course
165 uses `renv` (Ushey & Wickham, 2023) to simplify the installation process. Users simply run a
166 single command which installs the required packages for all successive R sessions. However,
167 `renv` only provides a simplified solution to reproducibility and can be mired by system
168 dependencies and versions of RStudio. Therefore, users can alternatively, pull a pre-made
169 Docker image building a container locally to host the RStudio environment. This does not
170 necessitate users to have R or RStudio installed - only Docker Desktop - maintaining a
171 more consistent and reproducible development environment across different systems and
172 platforms. In either case, generating the working environment requires minimal effort
173 (Figure 1.). Detailed guidance on how to recreate the working environment in both cases
174 are provided on the course website.

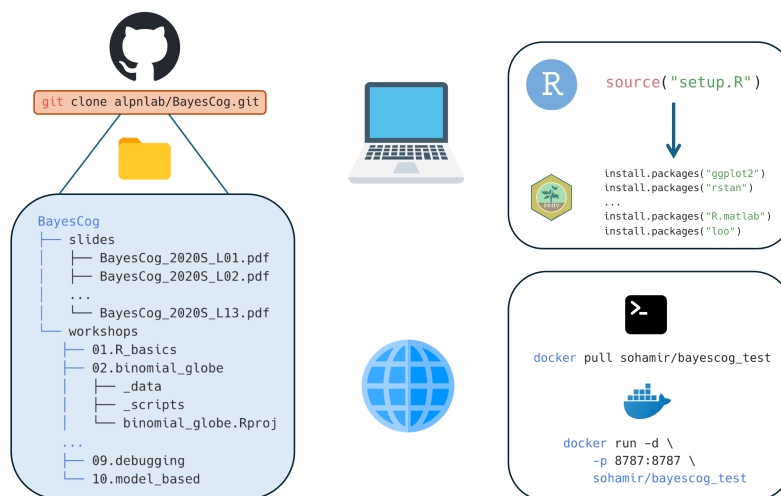


Figure 1. The course materials are hosted on GitHub and can be downloaded locally using the `git clone` command. Users have two options if wanting to replicate the working environment for the course materials. For working on one's own installation and version of RStudio, `renv` manages all required packages and dependencies. Conversely, if users would like to work on a specific RStudio version, they can pull the provided Docker image, which installs the required R packages on an RStudio server. In either case, the required dependencies are minimal (R/RStudio or Docker), and recreating the environment only involves running one or two simple commands. Icons from icons8.com

Future Directions

The BayesCog course provides a general introduction to Bayesian statistics and computational modeling within the context of psychological research. Subsequently, the materials and topics could easily be developed further. This includes using computational methods with neuroimaging data (Glascher & O'Doherty, 2010; Hollander et al., 2016), understanding behaviors in the social world (Kutlikova et al., 2023; Y. Pan et al., 2023), and the theory-based modeling of psychiatric disorders (Maia et al., 2017; Sohail & Zhang, 2024; Suter et al., 2025). Furthermore, the Stan programming language remains technically challenging, leading to the development of user-friendly packages for computational modeling (Ahn et al., 2017). Tutorials on how to implement these packages could further broaden the use of computational methods within the psychological sciences. In the era of large language models (LLMs) and generative AI, the audience are also encouraged to combine this course with LLM tools such as ChatGPT to explore a more tailored learning experience (Sohail & Lin, 2025; Sohail & Zhang, 2025).

To this end, we openly receive feedback and suggestions from the wider community. Broader comments can be communicated on the [GitHub repository](#) by either reporting an issue or requesting an enhancement. On-the-other-hand, we recommend major changes to be made communicated beforehand and – if appropriate - made directly by forking the repository and pushing changes to the main branch. A member of the contributing team will then review the changes. Accepted contributions will be credited and following the community guidelines outlined on the [CONTRIBUTORS](#) page.

Acknowledgments

A.S. would like to thank Lei Zhang for developing the initial version of these materials,

including slides, code, and GitHub repositories. A.S. and L.Z. would also like to acknowledge existing open-source materials, including Shawn Rhoads' Computational Models of Human Social Behavior and Neuroscience (Rhoads & Gan, 2022), Luke Chang's 'DartBrains: An online open access resource for learning functional neuroimaging analysis methods in Python' (Chang et al., 2020) and Magdalena Chechlac's 'MRI on BEAR' course in neuroimaging (Sohail et al., 2025), who influenced the development of these materials.

L.Z. is supported by the Wellcome Trust (228268/Z/23/Z) and Royal Society Royal Society (IES\R3\243253). A.S. is supported by an MRC AIM iCASE DTP Studentship (Ref: MR/W007002/1).

Author Contributions

L.Z. created, designed and taught the original course materials by developing the syllabus, writing the Stan and R code and creating and interpreting the datasets. A.S. created the website, adding the content by converting, editing and expanding the source material written by L.Z. L.Z. and A.S. revised the course materials and wrote the manuscript.

References

- Ahn, W.-Y., Haines, N., & Zhang, L. (2017). Revealing Neurocomputational Mechanisms of Reinforcement Learning and Decision-Making With the hBayesDM Package. *Computational Psychiatry (Cambridge, Mass.)*, 1, 24–57.
- Annis, J., & Palmeri, T. J. (2018). Bayesian statistical approaches to evaluating cognitive models. *Wiley Interdisciplinary Reviews: Cognitive Science*, 9(2), e1458.
- Baribault, B., & Collins, A. G. E. (2023). Troubleshooting Bayesian cognitive models. *Psychological Methods*.
- Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M. A., Guo, J., Li, P., & Riddell, A. (2017). Stan: A Probabilistic Programming Language. *Journal of Statistical Software*, 76, 1.
- Chang, L., Huckins, J., Cheong, J., Brietzke, S., Lindquist, M., & Wager, T. (2020). *DartBrains: An online open access resource for learning functional neuroimaging analysis methods in python*. Zenodo. <https://doi.org/10.5281/zenodo>.
- Crawley, D., Zhang, L., Jones, E. J., Ahmad, J., Oakley, B., San José Cáceres, A., Charman, T., Buitelaar, J. K., Murphy, D. G., Chatham, C., & others. (2020). Modeling flexible behavior in childhood to adulthood shows age-dependent learning mechanisms and less optimal learning in autism in each age group. *PLoS Biology*, 18(10), e3000908.
- Daw, N. D., Gershman, S. J., Seymour, B., Dayan, P., & Dolan, R. J. (2011). Model-based influences on humans' choices and striatal prediction errors. *Neuron*, 69(6), 1204–1215.
- Dayan, P., & Niv, Y. (2008). Reinforcement learning: The good, the bad and the ugly. *Current Opinion in Neurobiology*, 18(2), 185–196.
- Eckstein, M. K., Wilbrecht, L., & Collins, A. G. (2021). What do reinforcement learning models measure? Interpreting model parameters in cognition and neuroscience. *Current Opinion in Behavioral Sciences*, 41, 128–137.
- Farrell, S., & Lewandowsky, S. (2018). *Computational Modeling of Cognition and Behavior*. Cambridge University Press.
- FeldmanHall, O., & Nassar, M. R. (2021). The computational challenge of social learning. *Trends in Cognitive Sciences*, 25(12), 1045–1057.

- 250 Glascher, J. P., & O'Doherty, J. P. (2010). Model-based approaches to neuroimaging:
251 Combining reinforcement learning theory with fMRI data. *WIREs Cognitive Science*,
252 1(4), 501–510.
- 253 Guest, O., & Martin, A. E. (2021). How Computational Modeling Can Force The-
254 ory Building in Psychological Science. *Perspectives on Psychological Science*, 16(4),
255 789–802.
- 256 Hauser, T. U., Skvortsova, V., Choudhury, M. D., & Koutsouleris, N. (2022). The promise
257 of a model-based psychiatry: Building computational models of mental ill health. *The*
258 *Lancet Digital Health*, 4(11), e816–e828.
- 259 Hollander, G. de, Forstmann, B. U., & Brown, S. D. (2016). Different Ways of Linking Be-
260 havioral and Neural Data via Computational Cognitive Models. *Biological Psychiatry:*
261 *Cognitive Neuroscience and Neuroimaging*, 1(2), 101–109.
- 262 Huys, Q. J. M., Maia, T. V., & Frank, M. J. (2016). Computational psychiatry as a bridge
263 from neuroscience to clinical applications. *Nature Neuroscience*, 19(3), 404–413.
- 264 Kruschke, J. (2014). *Doing Bayesian Data Analysis: A Tutorial with R, JAGS, and Stan*.
265 Academic Press.
- 266 Kutlikova, H. H., Zhang, L., Eisenegger, C., Honk, J. van, & Lamm, C. (2023). Testosterone
267 eliminates strategic prosocial behavior through impacting choice consistency in healthy
268 males. *Neuropsychopharmacology*, 48(10), 1541–1550.
- 269 Lambert, B. (2018). *A Student's Guide to Bayesian Statistics*. SAGE.
- 270 Lee, D., Seo, H., & Jung, M. W. (2012). Neural basis of reinforcement learning and
271 decision making. *Annual Review of Neuroscience*, 35(1), 287–308.
- 272 Lee, M. D. (2011). How cognitive modeling can benefit from hierarchical Bayesian models.
273 *Journal of Mathematical Psychology*, 55(1), 1–7.
- 274 Lee, M. D., & Wagenmakers, E.-J. (2014). *Bayesian Cognitive Modeling: A Practical*
275 *Course*. Cambridge University Press.
- 276 Lockwood, P. L., & Klein-Flügge, M. C. (2021). Computational modelling of social
277 cognition and behaviour-a reinforcement learning primer. *Social Cognitive and Affective*
278 *Neuroscience*, 16(8), 761–771.
- 279 Maia, T. V., Huys, Q. J. M., & Frank, M. J. (2017). Theory-Based Computational
280 Psychiatry. *Biological Psychiatry*, 82(6), 382–384.
- 281 Marr, D. (1982). *Vision: A computational investigation into the human representation*
282 *and processing of visual information*. W.H. Freeman; Company.
- 283 McElreath, R. (2018). *Statistical Rethinking: A Bayesian Course with Examples in R and*
284 *Stan*. Chapman; Hall/CRC.
- 285 Niv Lab. (2021). *NivStan*. <https://nivlab.github.io/nivstan/>
- 286 Niv, Y. (2009). Reinforcement learning in the brain. *Journal of Mathematical Psychology*,
287 53(3), 139–154.
- 288 O'Reilly, J. X., & Ouden, E. M. den, Hanneke. (2015). *Models of learning*.
- 289 Palminteri, S., Wyart, V., & Koechlin, E. (2017). The Importance of Falsification in
290 Computational Cognitive Modeling. *Trends in Cognitive Sciences*, 21(6), 425–433.
- 291 Pan, W., Geng, H., Zhang, L., Fengler, A., Frank, M., Zhang, R., & Chuan-Peng, H.
292 (2022). A hitchhiker's guide to bayesian hierarchical drift-diffusion modeling with
293 docker-HDDM. *PsyArXiv*, 10.
- 294 Pan, Y., Vinding, M. C., Zhang, L., Lundqvist, D., & Olsson, A. (2023). A brain-to-

- 295 brain mechanism for social transmission of threat learning. *Advanced Science*, 10(28),
296 2304037.
- 297 Plummer, M., & others. (2003). *JAGS: A program for analysis of bayesian graphical*
298 *models using gibbs sampling*. 1–10.
- 299 Press, C., Yon, D., & Heyes, C. (2022). Building better theories. *Current Biology*, 32(1),
300 R13–R17.
- 301 Rescorla, R. A., & Wagner, A. R. (1972). *A theory of pavlovian conditioning: Variations*
302 *in the effectiveness of reinforcement and nonreinforcement* (pp. 64–99). Appleton-
303 Century-Crofts.
- 304 Rhoads, S. A., & Gan, L. (2022). Computational models of human social behavior and
305 neuroscience: An open educational course and jupyter book to advance computational
306 training. *Journal of Open Source Education*, 5(47), 146.
- 307 Rocca, R., & Yarkoni, T. (2021). Putting Psychology to the Test: Rethinking Model
308 Evaluation Through Benchmarking and Prediction. *Advances in Methods and Practices*
309 *in Psychological Science*, 4(3), 25152459211026864.
- 310 RStudio Team. (2020). *RStudio: Integrated development environment for r*.
- 311 Sohail, A., & Lin, Z. (2025). Recalibrating academic expertise in the age of generative AI.
312 Available at SSRN 5224452.
- 313 Sohail, A., Quinn, A. J., Carpenter, J. J., Hansen, P. C., & Chechacz, M. (2025). *MRI on*
314 *BEAR (birmingham environment for academic research): Magnetic resonance imaging*
315 *in cognitive neuroscience at the centre for human brain health* (Version v1.0.0). Zenodo.
316 <https://doi.org/10.5281/zenodo.15804715>
- 317 Sohail, A., & Zhang, L. (2024). Informing the treatment of social anxiety disorder with
318 computational and neuroimaging data. *Psychoradiology*, 4, kkae010.
- 319 Sohail, A., & Zhang, L. (2025). Using large language models to facilitate academic work
320 in the psychological sciences. *Current Psychology*, 44(9), 7910–7918.
- 321 Stan Development Team. (2024). *RStan: The R interface to Stan*.
- 322 Suter, G., Černis, E., & Zhang, L. (2025). Interpersonal computational modelling of social
323 synchrony in schizophrenia and beyond. *Psychoradiology*, 5, kkaf011.
- 324 Ushey, K., & Wickham, H. (2023). Renv: Project environments. *R Package Version*, 1(2).
- 325 Vehtari, A., Gabry, J., Magnusson, M., Yao, Y., Bürkner, P.-C., Paananen, T., & Gelman,
326 A. (2015). Loo: Efficient leave-one-out cross-validation and WAIC for bayesian models.
327 (*No Title*).
- 328 Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New
329 York.
- 330 Wilson, R. C., & Collins, A. G. (2019). Ten simple rules for the computational modeling
331 of behavioral data. *eLife*, 8, e49547.
- 332 Zhang, L., Lengersdorff, L., Mikus, N., Glascher, J., & Lamm, C. (2020). Using reinforce-
333 ment learning models in social neuroscience: Frameworks, pitfalls and suggestions of
334 best practices. *Social Cognitive and Affective Neuroscience*, 15(6), 695–707.